Lab Handout 1-D

Two-Node Packet Transmission and Reception

Capstone — Module 1: Radio Frequency (RF) Hardware Foundations

1 Objective

Transmit a known payload from one STM32+RFM95W node (Node A) and verify reception with correct Cyclic Redundancy Check (CRC), Received Signal Strength Indicator (RSSI), and Signal-to-Noise Ratio (SNR) readings on a second node (Node B). This is the first time the full RF link is exercised end-to-end and the baseline against which all Module 1 link-budget assumptions are validated.

2 Prerequisites

- Lab Handouts 1-A, 1-B, 1-C complete for at least **two** nodes.

3 Materials

- $2\times$ assembled RFM95W breakouts.
- $2\times$ Nucleo-L476RG boards + USB cables.
- $2\times$ host USB ports (or a powered USB hub).
- Breadboards and jumpers for both nodes.

4 Procedure

4.1 Step 1 — Configure LoRa modem parameters

Both nodes must use identical physical-layer parameters. Use Table 1 for this handout; these are the Module 1 reference settings and will also be used in Lab Handouts 1-F and 1-G.

Table 1: Baseline LoRa physical-layer configuration.

Parameter	**Value**	**Register(s)**
Carrier frequency	915.0 MHz	`RegFrfMsb/Mid/Lsb`
Spreading factor	SF7	`RegModemConfig2[7:4]`
Bandwidth	125 kHz	`RegModemConfig1[7:4]`
Coding rate	4/5	`RegModemConfig1[3:1]`
Header mode	Explicit	`RegModemConfig1[0]`
CRC	On	`RegModemConfig2[2]`
Preamble length	8 symbols	`RegPreambleMsb/Lsb`
Sync word	`0x34` (private)	`RegSyncWord`
Output power	+14 dBm (PA_BOOST)	`RegPaConfig`
Payload length (`TX`)	16 bytes	`RegPayloadLength`

4.2 Step 2 — Node A transmitter firmware

1. Starting from your Lab Handout 1-C project, add an initialization routine that programs all registers in Table 1, then leaves the modem in `STANDBY`.

2. In the main loop:

   (a) Build a 16-byte test payload. Bytes 0–3: ASCII `"TEST"`. Bytes 4–7: little-endian 32-bit sequence counter that increments each transmission. Bytes 8–15: arbitrary pattern (e.g., `0xDE 0xAD 0xBE 0xEF 0xCA 0xFE 0xBA 0xBE`).

   (b) Write the payload to the `FIFO` at `FifoTxBaseAddr`.

   (c) Set `RegPayloadLength = 16`.

   (d) Command `TX` mode (`RegOpMode = 0x83`).

   (e) Poll `RegIrqFlags` for `TxDone` (bit 3). Clear by writing `1` to that bit.

   (f) `printf` the sequence number over the virtual COM port.

   (g) Delay 1000 ms.

4.3 Step 3 — Node B receiver firmware

1. Use the same initialization, but set `FifoRxBaseAddr` and put the modem into continuous RX mode (`RegOpMode = 0x85`).
2. In the main loop (or in the `DIO0` interrupt, if you're ready), poll `RegIrqFlags` for `RxDone` (bit 6).
3. When `RxDone` asserts:
   (a) Check `PayloadCrcError` (bit 5). If set, increment a CRC-error counter and skip the payload; if clear, read the payload.
   (b) Read `RegRxNbBytes` for the received payload length.
   (c) Read `RegFifoRxCurrentAddr`, write it to `RegFifoAddrPtr`, then read `RxNbBytes` bytes from `RegFifo`.
   (d) Read `RegPktRssiValue` and convert: RSSI[dBm] = raw − 157 (high-frequency port).
   (e) Read `RegPktSnrValue` and convert: SNR[dB] = raw/4 (signed byte).
   (f) `printf` the sequence number, RSSI, and SNR.
   (g) Clear `RegIrqFlags` by writing `0xFF`.

4.4 Step 4 — Run the link

1. Plug both nodes into the host PC.
2. Open two serial monitor instances (one per virtual COM port).
3. Flash Node B (receiver) first and confirm it prints an “awaiting packets” banner.
4. Flash Node A (transmitter) and confirm it starts incrementing the sequence counter.
5. Place the nodes approximately 1 m apart on the bench with antennas upright.

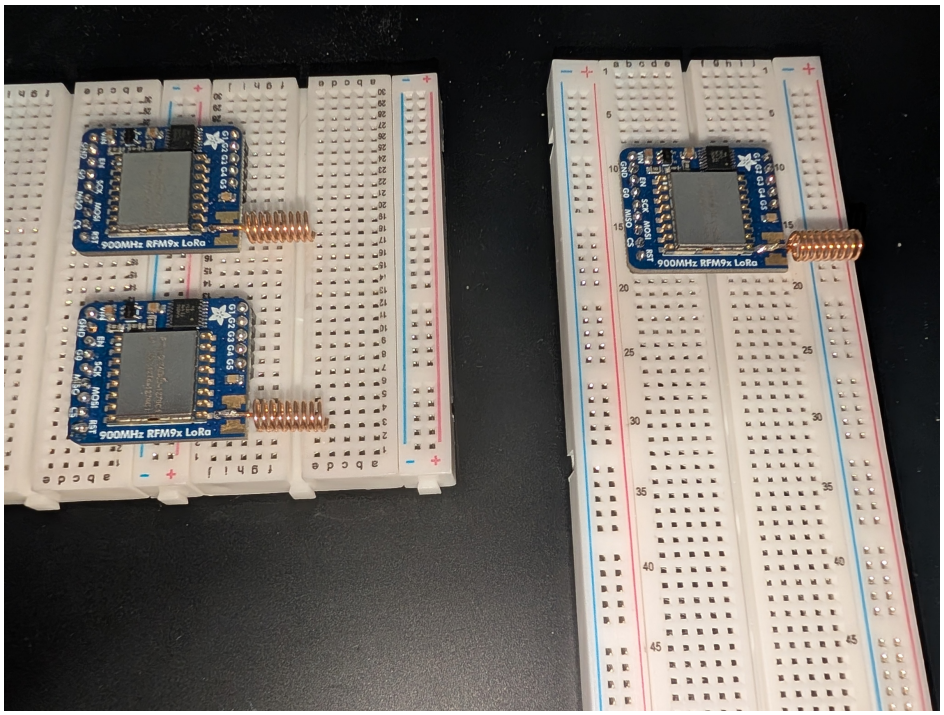

Figure 1: Both Nucleo+RFM95W nodes on the bench ≈1 m apart with spring antennas upright. Node A (TX) on the left, Node B (RX) on the right.

[Figure placeholder]

`LH1D_dual_serial.png`

Side-by-side serial monitors: Node A incrementing the sequence counter (left) and Node B printing matching sequence numbers with RSSI/SNR (right).

Figure 2: Side-by-side serial monitors: Node A incrementing the sequence counter (left) and Node B printing matching sequence numbers with RSSI/SNR (right).

4.5 Step 5 — Log 100 packets and analyze

1. Capture 100 consecutive receptions on Node B (copy from the serial monitor or pipe to a file).
2. Compute: packet reception rate (received / transmitted), mean RSSI, standard deviation of RSSI, mean SNR, and the count of CRC errors.
3. Compare mean RSSI against the Module 1 link budget estimate for 1 m indoor separation.

[Figure placeholder]

`LH1D_rssi_histogram.png`

Example RSSI histogram across 100 consecutive receptions at 1 m bench separation. Students generate their own version from the captured log as part of the deliverable.

Figure 3: Example RSSI histogram across 100 consecutive receptions at 1 m bench separation. Students generate their own version from the captured log as part of the deliverable.

5 Verification Checklist

- ☐ ≥95% packet reception rate at 1 m indoor line-of-sight.
- ☐ Zero CRC errors across 100 consecutive packets.
- ☐ Sequence counter values are contiguous on Node B (no gaps).
- ☐ Mean RSSI is within ±6 dB of the computed link budget.
- ☐ Mean SNR is >+5 dB.

Troubleshooting

- **No packets received**: check that both nodes use the same sync word and spreading factor. A one-bit mismatch yields silent failure.
- **High CRC error rate**: verify `CrcOn` in `RegModemConfig2` on both nodes, and confirm the preamble length matches.
- **Very low RSSI (< −90 dBm at 1 m)**: usually an antenna problem — re-check the spring antenna solder joint.
- **`TxDone` never asserts**: `DIO0` mapping wrong, or the modem is in FSK mode instead of LoRa (bit 7 of `RegOpMode`).

Lab Handout 1-D Deliverable

1. Source code for both Node A (TX) and Node B (RX).
2. Raw serial log of 100 consecutive receptions on Node B.
3. Summary statistics: reception rate, mean RSSI, RSSI standard deviation, mean SNR, CRC error count.
4. One-paragraph comparison between measured mean RSSI and the Module 1 link-budget prediction. Explain any discrepancy >6 dB.

Glossary

CRC	Cyclic Redundancy Check	1, 2, 3, 5, 6
RF	Radio Frequency	1
RSSI	Received Signal Strength Indicator	1, 3, 5, 6
SNR	Signal-to-Noise Ratio	1, 3, 5, 6